

ECO387C - Fall 2024

Discussion 12

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1 Aiyagari (1994)

Aiyagari (1994) model is foundational in heterogeneous agents literature in macroeconomics.¹ It is used for analyzing wealth inequality, and household consumption behavior under incomplete markets with idiosyncratic risk. Compared to Huggett (1993), the model features a production sector that allows for an endogenous non-zero, positive aggregate saving.² The model offers insights into wealth distribution and the impact of idiosyncratic income risk on savings behavior.

1.1 Households

The model considers a continuum of households on the unit interval $([0,1])$ that face idiosyncratic labor endowment shocks. Each household has a utility function over consumption and seeks to maximize lifetime utility:

$$\max_{\{c_{it}, k_{it+1}\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_{it})$$

Each household $i \in [0,1]$ receives an exogenous labor endowment n_{it} that follows a stochastic process. They supply this labor and receive wages. There is also a saving vehicle, capital with a rental rate r . Their budget constraint is given by

$$c_{it} + k_{it+1} = (1 + r - \delta)k_{it} + wn_{it}, \quad k_{it+1} \geq 0 \quad \forall t, \quad k_{i0} > 0 \text{ is given.}$$

where δ is the depreciation rate of capital. Note that households are identical in their preferences, the stochastic process governing the labor endowment, and the prices (rental

¹Aiyagari, S.R., 1994. Uninsured idiosyncratic risk and aggregate saving. The Quarterly Journal of Economics, 109(3), pp.659-684.

²Huggett, M., 1993. The risk-free rate in heterogeneous-agent incomplete-insurance economies. Journal of Economic Dynamics and Control, 17(5-6), pp.953-969.

rate and wage) they face. However, the households differ in their *realization of labor endowment* as well as their initial capital. In other words, they face the same risk process, but different realization. The aggregate supply of capital and labor are

$$K_t^s = \int_0^1 k_{it} di \quad N_t^s = \int_0^1 n_{it} di$$

The superscript s indicates that the households are the *suppliers* of the factors of production. K_t^s is the aggregate saving in the economy at time t . The law of motion for capital is

$$K_{t+1}^s = I_t + (1 - \delta)K_t^s$$

where $I_t = \int_0^1 I_{it} di = \int_0^1 k_{it+1} - (1 - \delta)k_{it} di$ is aggregate investment.

1.2 Firm

The production side consists of a representative firm operating under perfect competition. The firm has a standard Cobb-Douglas production function:

$$Y_t = (K_t^d)^\alpha (N_t^d)^{1-\alpha}$$

where Y_t is output, K_t and N_t are *aggregate* capital and labor. α is the capital share. The firm hires labor and borrows capital from households, paying wages w and a rental rate r , which are determined in equilibrium. The superscript d indicates that the firm is the *demand*er of these factors of production. The firms take these prices as given when solving their profit maximization problem:

$$\max_{K_t^d, N_t^d} (K_t^d)^\alpha (N_t^d)^{1-\alpha} - wN_t^d - rK_t^d$$

The first-order conditions of the firm's problem result in factor demand conditions:

$$r = \alpha(K_t^d)^{\alpha-1} (N_t^d)^{1-\alpha} \quad w = (1 - \alpha)(K_t^d)^\alpha (N_t^d)^{-\alpha}$$

1.3 Market Clearing

In equilibrium, there are three markets to be cleared out:

1. Goods market: The aggregate output produced by the firm should be either consumed or invested (by saving in capital) by households.

$$Y_t = C_t + I_t$$

where $C_t = \int_0^1 c_{it} di$ and $I_t = \int_0^1 I_{it} di = \int_0^1 k_{it+1} - (1 - \delta)k_{it} di$

2. Capital market: The demand for capital by the firm should be equated with the aggregate supply of capital from the households.

$$K_t^d = K_t^s = \int_0^1 k_{it} di$$

3. Labor market: The demand for labor by the firm should be equated with the aggregate supply of labor from the households.

$$N_t^d = N_t^s = \int_0^1 n_{it} di$$

The prices (r, w) should adjust such that they clear out these markets while satisfying the optimality of households and the representative firm.

2 Computation

1. Discretize the process for labor endowment and compute the aggregate labor using the law of large numbers.
2. Set a fixed value of $K = K_j$ with $j = 0$, compute (w, r) from the FOCs of the firm.
3. Solve the household's optimization problem using value function iteration and obtain policy functions.
4. Use the optimal policy $a'_j = g_j(k, n)$ to deduce an associated stationary distribution $\lambda_j(k, n)$.
5. Compute the average value of capital associated with $\lambda_j(k, n)$

$$K_j^* = \sum_{k, n} \lambda_j(k, n) g_j(k, n).$$

6. For a fixed "relaxation parameter" $\eta \in (0, 1)$, compute a new estimate of K from the method

$$K_{j+1} = \eta K_j + (1 - \eta) K_j^*.$$

7. Iterate on this scheme to convergence.